Dhirubhai Ambani Institute of Information and Communication Technology (DA-IICT)

Discrete Mathematics SC612

IN SEMESTER 1 EXAMINATION

Max Marks - 50 | Autumn 2023-24 | Time - 1:30 hours

- All questions are compulsory.
- Answers must have proper justification to get full credit.
- Understanding the questions is part of the evaluation. Don't ask for any clarification. In case of ambiguity, proceed with an assumption by clearly stating it.

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1. Find the logical expressions for the given English statements using predicates, quantifiers, logical connectives and appropriately defined propositional variables/functions.

   (a) It is not safe to hike on the trail, but grizzly bears have not been seen in the area and the berries along the trail are ripe. [02]

   (b) For hiking on the trail to be safe, it is necessary but not sufficient that berries not be ripe along the trail and for grizzly bears not to have been seen in the area. [02]

   (c) There exists a student who has taken a mathematics course, and for every other student who has taken a mathematics course, there exists a student who has taken a computer science course. [03]

   (d) For every student who has taken a computer science course, there exists a different student who has not taken a mathematics course. [03]

Solution: Let p, q, and r be the propositions

p : Grizzly bears have been seen in the area.

q : Hiking is safe on the trail.

r : Berries are ripe along the trail.

(a) It is not safe to hike on the trail: $\neg q$
grizzly bears have not been seen in the area: $\neg p$
Therefore, $\neg q \wedge \neg p \wedge r$

(b) Berries not be ripe along the trail and grizzly bears not to have been seen in the area: $\neg p \wedge \neg r$
For hiking on the trail to be safe, it is **necessary** that berries not be ripe along the trail and for grizzly bears not to have been seen in the area: $q \rightarrow (\neg p \wedge \neg r)$

For hiking on the trail to be safe, **not sufficient** that berries not be ripe along the trail and for grizzly bears not to have been seen in the area: $\neg(\neg p \wedge \neg r) \rightarrow q$
Therefore, $(q \rightarrow (\neg p \wedge \neg r)) \wedge (\neg(\neg p \wedge \neg r) \rightarrow q)$

(c) Let $M(x)$: Student x has taken a mathematics course
$C(x)$: Student x has taken a computer science course.

There exists a student who has taken a mathematics course: $\exists x M(x)$

Every other student who has taken a mathematics course, there exists a student who has taken a computer science course:$\forall y(M(y) \rightarrow \exists z(C(z) \wedge x \neq z))$

Therefore, $\exists x(M(x) \wedge \forall y(M(y) \rightarrow \exists z(C(z) \wedge x \neq z)))$

(d) Let $M(x)$: Student x has taken a mathematics course
$C(x)$: Student x has taken a computer science course.

Every student who has taken a computer science course: $\forall x C(x)$

There exists a different student who has not taken a mathematics course:$\exists y(\neg M(y) \wedge x \neq y)$

Therefore, $\forall x(C(x) \rightarrow \exists y(\neg M(y) \wedge x \neq y))$

2. Without using a truth table, determine all the truth values of p, q, r and s for which the logical expression $(p \vee q \vee r \vee s) \wedge (\neg p \vee \neg q \vee \neg r \vee \neg s)$ will be true? Justify your answer. [04]

Solution: Claim: To determine the truth values of p, q, r, and s for which the logical expression $(p \vee q \vee r \vee s) \wedge (\neg p \vee \neg q \vee \neg r \vee \neg s)$ will be true.

The given expression is a conjunction (AND) of two sub-expressions: $(p \vee q \vee r \vee s)$ and $(\neg p \vee \neg q \vee \neg r \vee \neg s)$. For the conjunction to be true, both of these sub-expressions must be true.

1. The first sub-expression $(p \vee q \vee r \vee s)$ is true if at least one of p, q, r, or s is true.
2. The second sub-expression $(\neg p \vee \neg q \vee \neg r \vee \neg s)$ is true if at least one of $\neg p, \neg q, \neg r$, or $\neg s$ is true. This is equivalent to saying that none of p, q, r, or s are true.

To summarize, the entire expression $(p \vee q \vee r \vee s) \wedge (\neg p \vee \neg q \vee \neg r \vee \neg s)$ is true if and only if at least one of p, q, r, or s is true, and none of them are true simultaneously. This is the definition of an exclusive OR (XOR) relationship.

So, the truth values of p, q, r, and s for which the expression is true are:

Exactly one of p, q, r, or s is true, and the rest are false.

Thus, the composite statement to be true, p,q,r,s shouldn't all be same.

Exclude the cases of all T and all F.

Checking

p	q	r	s	$r \vee s$	$q \vee r \vee s$	$p \vee q \vee r \vee s$	$\neg p$	$\neg q$	$\neg r$	$\neg s$	$\neg r \vee \neg s$	$\neg q \vee \neg r \vee \neg s$	$\neg p \vee \neg q \vee \neg r \vee \neg s$	$(p \vee q \vee r \vee s) \wedge (\neg p \vee \neg q \vee \neg r \vee \neg s)$
T	T	T	T	T	T	T	F	F	F	F	F	F	F	F
T	T	T	F	T	T	T	F	F	F	T	T	T	T	T
T	T	F	T	T	T	T	F	F	T	F	T	T	T	T
T	T	F	F	F	T	T	F	F	T	T	T	T	T	T
T	F	T	T	T	T	T	F	T	F	F	F	T	T	T
T	F	T	F	T	T	T	F	T	F	T	T	T	T	T
T	F	F	T	T	T	T	F	T	T	F	T	T	T	T
T	F	F	F	F	F	T	F	T	T	T	T	T	T	T
F	T	T	T	T	T	T	T	F	F	F	F	F	T	T
F	T	T	F	T	T	T	T	F	F	T	T	T	T	T
F	T	F	T	T	T	T	T	F	T	F	T	T	T	T
F	T	F	F	F	T	T	T	F	T	T	T	T	T	T
F	F	T	T	T	T	T	T	T	F	F	F	T	T	T
F	F	T	F	T	T	T	T	T	F	T	T	T	T	T
F	F	F	T	T	T	T	T	T	T	F	T	T	T	T
F	F	F	F	F	F	F	T	T	T	T	T	T	T	F

3. Construct a truth table for the compound proposition $(\neg p \leftrightarrow \neg q) \leftrightarrow (q \leftrightarrow r)$ [04]

Solution:

p	q	r	$\neg p$	$\neg q$	$\neg p \leftrightarrow \neg q$	$q \leftrightarrow r$	$(\neg p \leftrightarrow \neg q) \leftrightarrow (q \leftrightarrow r)$
T	T	T	F	F	T	T	T
T	T	F	F	F	T	F	F
T	F	T	F	T	F	F	T
T	F	F	F	T	F	T	F
F	T	T	T	F	F	T	F
F	T	F	T	F	F	F	T
F	F	T	T	T	T	F	F
F	F	F	T	T	T	T	T

4. Check if $(p \to q) \to (r \to s)$ and $(p \to r) \to (q \to s)$ are logically equivalent? [04]

Solution: Let p, q, r, s have respectively, F, F, T and F truth values.

1. Now, $p \to q = F \to F = T$ and $r \to s = T \to F = F$
   $\therefore (p \to q) \to (r \to s) = T \to F = F$

2. Again, $p \to r = F \to T = T$ and $q \to s = F \to F = T$
$\therefore (p \to r) \to (q \to s) = T \to T = T$

We found one assignment of truth values to p, q, r, and s for which the truth values of $(p \to q) \to (r \to s)$ and $(p \to r) \to (q \to s)$ differ.

Hence, both are not logically equivalent.

Alternate method

p	q	r	s	$p \to q$	$r \to s$	$\boldsymbol{(p \to q) \to (r \to s)}$	$p \to r$	$q \to s$	$\boldsymbol{(p \to r) \to (q \to s)}$
T	T	T	T	T	T	T	T	T	T
T	T	T	F	T	F	F	T	F	F
T	T	F	T	T	T	T	F	T	T
T	T	F	F	T	T	T	F	F	T
T	F	T	T	F	T	T	T	T	T
T	F	T	F	F	F	T	T	T	T
T	F	F	T	F	T	T	F	T	T
T	F	F	F	F	T	T	F	T	T
F	T	T	T	T	T	T	T	T	T
F	T	T	F	T	F	F	T	F	F
F	T	F	T	T	T	T	T	T	T
F	T	F	F	T	T	$\boldsymbol{T}$	T	F	$\boldsymbol{F}$
F	F	T	T	T	T	T	T	T	T
F	F	T	F	T	F	F	T	T	T
F	F	F	T	T	T	T	T	T	T
F	F	F	F	T	T	T	T	T	T

5. Consider the following premises:

   1. John stays at home unless it is not raining.
   2. John stays at home only if he watches television.
   3. John goes for a walk whenever it is not raining.
   4. John either watches television or goes for a walk.

   Use Rules of Inference to determine whether or not John goes for a walk. [06]

Solution: Let us consider the logical form for the above propositions as

h: John stays at home

r: It is raining.

t: John watches television.

w: John goes for a walk

The premises are now

1. h unless $\neg r$ reduces to $r \to h$
2. h only if t reduces to $h \to t$
3. w whenever $\neg r$ reduces to $\neg r \to w$
4. either t or w reduces to $t \vee w$

$\therefore$ whether or not w

5. Using Hypothetical syllogism on 1 & 2 $r \to t$ ($\equiv \neg t \to \neg r$)
6. $\neg r \to w$
7. Hypo syll on 5 & 6 $\neg t \to w$ ($\equiv t \vee w$)
8. 4 and 7 are the same.

No conclusion

r	h	t	w	$r \to h$	$h \to t$	$\neg r$	$\neg r \to w$	$t \vee w$	$(\neg r \to w) \wedge (t \vee w)$	$(h \to t) \wedge (\neg r \to w) \wedge (t \vee w)$	$(r \to h) \wedge (h \to t) \wedge (\neg r \to w) \wedge (t \vee w)$	$(r \to h) \wedge (h \to t) \wedge (\neg r \to w) \wedge (t \vee w) \to w$
T	T	T	T	T	T	F	T	T	T	T	T	T
T	T	T	F	T	T	F	T	T	T	T	T	F
T	T	F	T	T	F	F	T	T	T	F	F	T
T	T	F	F	T	F	F	T	F	F	F	F	T
T	F	T	T	F	T	F	T	T	T	T	F	T
T	F	T	F	F	T	F	T	T	T	T	F	T
T	F	F	T	F	T	F	T	T	T	T	F	T
T	F	F	F	F	T	F	T	F	F	F	F	T
F	T	T	T	T	T	T	T	T	T	T	T	T
F	T	T	F	T	T	T	F	T	F	F	F	T
F	T	F	T	T	F	T	T	T	T	F	F	T
F	T	F	F	T	F	T	F	F	F	F	F	T
F	F	T	T	T	T	T	T	T	T	T	T	T
F	F	T	F	T	T	T	F	T	F	F	F	T
F	F	F	T	T	T	T	T	T	T	T	T	T
F	F	F	F	T	T	T	F	F	F	F	F	T

6. Determine whether the argument is correct or incorrect and explain why. [04]

Every computer science major takes discrete mathematics.
Natasha is taking discrete mathematics.
Therefore, Natasha is a computer science major.

Solution: Let $C(x)$ represents x is a computer science major.
$D(x)$ represents x takes discrete mathematics.

Premise 1: Every computer science major takes discrete mathematics.

$\forall x$, if x is a computer science major $C(x)$, then x takes discrete mathematics $D(x)$.

Premise 2: Natasha is taking discrete mathematics. $D(\text{Natasha})$

$\forall x(C(x) \rightarrow D(x))$
$C(a) \rightarrow D(a)$ (by universal instantiation)
$D(a)$

$\therefore\ C(a)$

This is invalid. As, it contains the fallacy of affirming the conclusion.

7. (a) Prove or disprove that the sum of an irrational number and a rational number is irrational. [04]

   (b) Using proof by contraposition, show that if n is an integer and $3n + 2$ is even, then n is even. [04]

Solution: (a) Let's prove that the sum of an irrational number and a rational number is irrational.

We will use Proof by Contradiction:

Assume, for the sake of contradiction, that the sum of an irrational number and a rational number is rational.

Let x be an irrational number, and y be a rational number.

Claim: To show that $x + y$ is irrational.

Assume that $x+y$ is rational. Then, by definition, there exist two integers m and $n \neq 0$ such that

$$x + y = m/n$$

and m and n have no common factors other than 1 (i.e., they are coprime).

$$x = \frac{m}{n} - y$$

RHS is the difference of two rationals. Therefore, x is a rational number.

However, this contradicts our initial assumption that x is an irrational number.

Thus, our assumption that $x + y$ is rational must be false.

Therefore, the sum of an irrational number and a rational number is indeed irrational.

This completes the proof by contradiction.

(b) To prove, If $3n + 2$ is even, then n is even.$\equiv p \rightarrow q$
We will use the Proof by contraposition($\neg q \rightarrow \neg p$): If n is odd, then $3n + 2$ is odd.

Assume that n is odd $\implies$ $n = 2k+1$ for some integer k.
Now, $3n+2 = 3(2k+1)+2 = 6k+5 = 2(3k+2)+1 = 2m+1$, where $m = 3k+2$ is an integer.
Thus $3n+2$ is odd.

8. Use mathematical induction to prove that $7^{n+2} + 8^{2n+1}$ is divisible by 57 for every nonnegative integer n. [05]

Solution: Let $P(n)$: "$7^{n+2} + 8^{2n+1}$ is divisible by 57" be the proposition.
Claim: To prove $P(n)$ is true for all $n \geq 0$

BASIS STEP: To show that $P(0)$ is true.
Now $P(0) := 7^{0+2} + 8^{2\cdot 0+1} = 7^2 + 8^1 = 57$ is divisible by 57 .

INDUCTIVE STEP: Assume that $P(k)$ is true for an arbitrary nonnegative integer k; that is, we assume that $7^{k+2} + 8^{2k+1}$ is divisible by 57.
To show that $P(k+1)$ **is true**, that is, $7^{(k+1)+2} + 8^{2(k+1)+1}$ is divisible by 57.

$$\begin{aligned} P(k+1) := 7^{(k+1)+2} + 8^{2(k+1)+1} &= 7^{k+3} + 8^{2k+3} \\ &= 7 \cdot 7^{k+2} + 8^2 \cdot 8^{2k+1} \\ &= 7 \cdot 7^{k+2} + 64 \cdot 8^{2k+1} \\ &= 7\left(7^{k+2} + 8^{2k+1}\right) + 57 \cdot 8^{2k+1} \\ &= 7 \cdot P(k) + 57 \cdot 8^{2k+1} \end{aligned}$$

The first term in the sum is divisible by 57 since $P(k)$ is divisible by 57 by Inductive hypothesis.
The second term in the sum, $57 \cdot 8^{2k+1}$, is divisible by 57.
Hence, $7 \cdot P(k) + 57 \cdot 8^{2k+1} = 7\left(7^{k+2} + 8^{2k+1}\right) + 57 \cdot 8^{2k+1} = 7^{k+3} + 8^{2k+3}$ is divisible by 57.

Because we have completed both the basis step and the inductive step, by the principle of mathematical induction we know that $7^{n+2}+8^{2n+1}$ is divisible by 57 for every nonnegative integer n.

9. Use strong induction to show that every positive integer n can be written as a sum of distinct powers of 2. [05]

Solution: Let $P(n)$: "$n > 0$ can be written as a sum of distinct powers of 2" be the proposition.
Claim: To prove $P(n)$ is true for all $n > 0$

BASIS STEP: To show that $P(1)$ is true.
Now $P(1) := 1 = 2^0$. The subsequent steps that $2 = 2^1, 3 = 2^1 + 2^0, 4 = 2^2$, $5 = 2^2 + 2^0$, and so on are true.

INDUCTIVE STEP: Assume the inductive hypothesis, $P(1), \ldots, P(k)$ are all true, that is, every positive integer up to k can be written as a sum of distinct powers of 2.

Claim: To show that $P(k+1)$ is true, that is, $k+1$ can be written as a sum of distinct powers of 2.

If $k+1$ is odd, then k is even, so 2^0 was not part of the sum for k.
Therefore the sum for $k+1$ is the same as the sum for k with the extra term 2^0 added. Therefore, $P(k+1)$ is true.

If $k+1$ is even, then $(k+1)/2$ is a positive integer, so by the inductive hypothesis $(k+1)/2$ can be written as a sum of distinct powers of 2.
$\frac{k+1}{2} = 2^{n_1} + 2^{n_2} + \cdots + 2^1$, $n_1, n_2, \ldots$, are distinct positive integers.
$k+1 = 2^{n_1+1} + 2^{n_2+1} + \cdots + 2^2$.
Therefore, $P(k+1)$ is true.

Hence, $P(n)$ is true for n.

******************************** END *********************************

This exam has 9 questions, for a total of 50 points.

Question:	1	2	3	4	5	6	7	8	9	Total
Points:	10	4	4	4	6	4	8	5	5	50
Score:										